INCH-POUND

MIL-DTL-43204D
29 April 2024

SUPERSEDING
MIL-C-43204C
31 December 1985

DETAIL SPECIFICATION

CLOTH, SPACER (OLEFIN)

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE.

1.1 Scope. This specification covers the requirements for the polypropylene mesh spacer cloth.

1.2 Classification. The spacer cloth needs to be of the following types as specified (see 6.2).

Type I - 9.5 oz per sq yd (min)
Type II - 15.0 oz per sq yd (min)
Type III - 8.25 oz per sq yd (min)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in Sections 3 and 4 of this detailed specification. This section does not include documents cited in other sections of this detailed specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in Sections 3 and 4 of this detailed specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to AFLCMC/EZSS, Bldg 28, 2145 Monahan Way, Wright-Patterson AFB OH 45433-7017 or emailed to engineering.standards@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2 Government documents.

2.3 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	Military Marking for Shipment and Storage

(Copies of this document are available online at https://assist.dla.mil.)

2.3.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of the document to the extent specified herein. Unless otherwise specified, the issue of these documents are those cited in the solicitation or contract.

FEDERAL TRADE COMMISSION SPECIFICATIONS

16 CFR Part 303	Rules and Regulations under the Textile Fiber Products Identification Act

(Copies of this document are available online at https://www.ftc.gov.)

2.4 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ) SPECIFICATIONS

ANSI/ASQ Z1.4	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://www.asq.org.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D-3775	Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics
ASTM-D-3776	Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM-D-5035	Standard Test Method for Breaking Force and Elongation of Textile Fabris (Strip Method)
ASTM-D-6193	Standard Practices for Stitches and Seams

(Copies of these documents are available online at https://www.astm.org.)

FEDERAL ACQUISITION REGULATION (FAR) SPECIFICATIONS

FAR 52.209	Qualification Requirements

(Copies of this document are available online at https://www.acquisition.gov.)

2.5 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1. First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3 and 6.3.

3.2. Materials.

3.2.1. Polyethylene yarn (Type I and Type III). – The polyethylene yarn for warp No. 1 and No. 2 shall be shrinkable branch type polyethylene. The yarn diameter shall be 0.010 inch ±0.001 inch initially, prior to fabrication of the finished cloth.

3.2.2. Polypropylene yarn (Type I and Type III). – The polypropylene yarn for warp No. 3 and the filling shall have a diameter of 0.010 inch ±0.001 inch initially, prior to fabrication of the finished cloth.

3.2.3. Polyethylene yarn (Type II). – The polyethylene yarn for the warp and filling shall be medium density polyethylene type. Yarn size for the warp and filling shall be 960 denier prior to finishing of the woven cloth.

3.3. Color. – The color of the cloth shall be that resulting from the color of the yarns available at the time of acquisition.

3.4. Physical requirements. – The finished cloth shall conform to the requirements specified in TABLE I when tested as specified in 4.5.

TABLE I. Physical requirements.

Characteristics		Requirements	
	TYPE I	TYPE II	TYPE III
Weight oz/sq yd	9.5 min,	15.0 min,	8.25 min,
	11.5 max	19.5 max	10.25 max
Thickness, min (inches)			
Initial	0.17	0.12	0.170
Compressed	0.075	0.07	0.045
After compression	0.15	0.1	0.15
Texture, min			
Warp No. 1 - Polyethylene	20 ends/inch	58 ends/inch	19 ends/inch
Warp No. 2 - Polyethylene	19 ends/inch	…	19 ends/inch
Warp No. 3 - Polypropylene	27 ends/inch	…	25 ends/inch
Filling - Polypropylene	70 picks/inches	…	64 picks/inch
- Polyethylene	…	35 picks/inch	…
Breaking strength, min (pounds)			
Warp	110	175	110
Filling	200	100	180

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3.4.1. Width. – Unless otherwise specified (see 6.2), the width shall be a minimum of 58 inches, inclusive of selvage.

3.4.2. Weave.

3.4.2.1. Type I and Type III. – The cloth shall have a weave that is formed by the weaving of three warps. The yarn for warp No. 1 and No. 2 shall conform to 3.2.1, and the yarn for warp No. 3 and the filling shall conform to 3.2.2. The weave pattern shall be as shown on FIGURE 1.

3.4.2.2. Type II. – The cloth shall have a honeycomb weave pattern as shown on FIGURE 2.

3.5. Finishing. – The cloth shall be so processed as to meet the physical requirements in 3.4 and the dimensional stability requirement in 3.5.1.

3.5.1. Dimensional stability.

3.5.1.1. Type I. – The finished Type I cloth shall show a dimensional change of not more than 4.0 percent in the warp direction and not more than 2.0 percent in the filling direction when tested as specified in 4.5.

3.5.1.2. Type II. – The finished Type II cloth shall show a dimensional change of not more than 8.5 percent in the warp direction and not more than 6.2 percent in the filling direction when tested as specified in 4.5.

3.5.1.3. Type III. – The finished Type III cloth shall show a dimensional change of not more than 8.5 percent in the warp direction and not more than 2.0 percent in the filling direction when tested as specified in 4.5.

3.6. Length and put-up. – Unless otherwise specified, the finished cloth shall be furnished in continuous lengths, each not less than 35 yards. The cloth shall be put up in a full width roll as specified in MIL-STD-129.

3.7. Fiber identification. – Each roll of cloth shall be labeled or ticketed for fiber content in accordance with 16 CFR Part 303.

3.8. Workmanship. – The finished cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. VERIFICATION

4.1. Classification of inspection. – The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.2. Responsibility for inspection. – Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2.1. Certificate of compliance. – Where certificates of compliance are submitted, the Government reserves the right to check-test such items to determine the validity of the certification.

4.3. First article inspection. – When a first article is required (see 6.2), it shall be visually inspected for appearance.

4.4. Quality conformance inspection. – Unless otherwise specified, sampling for inspection shall be performed in accordance with ANSI/ASQ Z1.4.

4.4.1. Component and material inspection. – In accordance with 4.2, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2. Examination of the end item. – Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.3.

4.4.2.1. Yard-by-yard examination. – The defects listed below shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only one defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The sample unit shall be 1 linear yard. The inspection level shall be I of the ANSI/ASQ Z1.4 inspection levels, and the acceptable quality level (AQL) shall be 6.5 defects per hundred yards. An approximately equal number of yards shall be examined from each roll selected. (The number of rolls from which the sample is to be selected shall be in accordance with TABLE II.)

Defects
Broken pick over one quarter the width of the cloth
Width less than specified
Harness skip
End out over three inches
Open or thin place
Wrong draw
Spots and stains
Any cut, hole, or tear
Mispick
Misweave

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4.4.2.2. Overall examination. – Each defect listed below shall be counted not more than once in each roll examined. The sample unit shall be one roll. The sample size (number of rolls selected as sample) and the number of defects acceptable shall be shown in TABLE II.

Defects
Overall uncleanness or soiled
Buckled, ridgy, wavy, or distorted

Table II. Sample Size.

Lot size in yards	Sample size in rolls	Acceptable number of defects
Up to 1200 1/	3	0
1201 up to and including 3200	5	0
3201 up to and including 10,000	8	0
10,001 up to and including 35,000	13	0
35,001 and over	20	1
1/ If lot contains fewer than 3 rolls, each roll in the lot shall be examined.		

4.4.2.3. Length examination.

4.4.2.3.1. Examination for length of individual rolls. – The rolls shall be examined for gross length. Any gross length found to be less than the specified minimum length or any gross length found to be more than 1 yard below the gross length marked on the roll ticket shall be considered as a defect with respect to length. The sample unit for this examination shall be one roll. The number of rolls selected as a sample (sample size) for this examination and the acceptance number shall be in accordance with TABLE II.

4.4.2.3.2. Examination for total yardage in sample. – The lot shall be unacceptable if the total of the actual gross length of rolls in the sample is less than the total of the gross lengths marked on roll tickets.

4.4.3. Packaging inspection. – The inspection shall be in accordance with the quality assurance provisions of MIL-STD-129.

4.5. Tests. – The methods of testing specified in ASTM-D-3776, ASTM-D-3775, and ASTM-D-5035, wherever applicable and as specified in TABLE III, shall be followed. The sample unit shall be 2 continuous yards full width of the finished cloth. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. The lot size shall be expressed in units of 1 yard. The sample size (number of sample units) shall be in accordance with the following:

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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Table III. Test methods.

Characteristic	Requirement reference	Test method
Yarn identification	3.2.1, 3.2.2, and 3.2.3	1/
Yarn diameter, denier	3.2.1, 3.2.2, and 3.2.3	1/
Weight	3.4	ASTM-D-3776
Thickness		
Initial	3.4	4.5.1
Compressed	3.4	4.5.1.3
After compression	3.4	4.5.1.3
Texture		
Warp No. 1 – Polyethylene	3.4	ASTM-D-3775
Warp No. 2 – Polyethylene	3.4	ASTM-D-3775
Warn No. 3 – Polypropylene	3.4	ASTM-D-3775
Filling – Polypropylene	3.4	ASTM-D-3775
Filling - Polyethylene	3.4	ASTM-D-3775
Breaking strength		
Warp	3.4	ASTM-D-5035
Filling	3.4	ASTM-D-5035
Weave	3.4.2	Visual 1/
Dimensional stability	3.5.1	4.5.2
1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.		

4.5.1. Determination of thickness values.

4.5.1.1. Preparation of specimen. – Five specimens shall be cut from different parts of a full width sample and shall measure not less than 6 inches by 6 inches.

4.5.1.2. Apparatus. – The test apparatus shall be a compressometer with a round pressure foot of 20 square inches area with a bearing surface capable of measuring 0.01 pound per square inch to 10.0 pounds per square inch without moving the specimen from under the pressure foot once the test is started. The pressure plate shall travel 0.2 inch per minute.

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4.5.1.3. Procedure. – Balance the compressometer. After balancing, place a test specimen under the pressure foot area and set the machine for 0.10 pound per square inch thickness. Lower the pressure foot and record the thickness reading under 0.10 pound per square inch as “Initial Thickness of Specimen”. Immediately after determining the initial thickness, lower the pressure foot until the pressure on the specimen has reached 10.0 pounds per square inch. Maintain the 10.0 pounds per square inch pressure on the specimen for one minute. At the end of the 1-minute period, record the thickness of the specimen as the “compressed Thickness”. Remove the pressure immediately and completely. After removal of the pressure, allow the specimen to relax for five minutes. Repeat the procedure of compression and relaxation five times on the same specimen. After the fifth load of 10 pounds has been removed and the specimen again relaxed without load for five minutes, measure the specimen at 0.10 pounds per square inch and read the thickness reading as the “Thickness of the Specimen after Compression”.

4.5.1.4. Report. – The initial thickness, compressed thickness and thickness after compression shall be determined on five specimens, and the individual characteristic results averaged and reported to the nearest 0.001 inch.

4.5.2. Dimensional stability.

4.5.2.1. Test specimen. – The specimen shall be a square of cloth 24 by 24 inches cut from different areas of the material and marked off in 18-inch lengths in the warp and filling direction. No specimen shall contain the same filling yarns.

4.5.2.2. Apparatus. – Hot air oven capable of maintaining a temperature of 200 ±5 °F.

4.5.2.3. Procedure. – Place the specimen in the hot air oven for 30 minutes at a temperature of 200 ±5 °F. Remove the specimen and allow it to cool before determining the dimensional changes of the cloth. Unless otherwise specified, three specimens shall be tested from each sample unit. The percent change of the sample unit shall be tested from each sample unit. The percent change of the sample unit shall be the average of the three specimens tested and shall be reported separately for the warp and filling to the nearest 0.1 percent. When a test result registers elongation (gain) rather than a shrinkage, each elongation result shall be prefixed with a minus sign and both the minus sign and the value enclosed in parentheses.

5. PACKAGING.

5.1. Packaging. – For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activities within the Military Service or Defense Agency, or within the military service’s system commands. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1. Intended use. – This cloth is intended to be used as a spacer fabric in anti-g garments.

6.2. Acquisition requirements. – Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type required (see 1.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Width of cloth required if other than specified (see 3.4.1).
e. Length and put-up required if other than specified (see 3.6).
f. Selection of applicable levels of put-up, preservation, and packing (see 5.1 and 5.2).

6.3. First article. – When a first article is required, it should be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The first Article should consist of 2 yards. The contracting officer should include specific instruction in all acquisition instruments regarding arrangements for examinations, test, and approval of the first article. (see 6.4 Shrinkage). The polypropylene is considered as being a nonshrinkable portion of the fabric, but basically polypropylene yarn will have up to 5 percent residual shrinkage.

6.4. Shrinkage. – The polypropylene is considered as being a nonshrinkable portion of the fabric, but basically polypropylene yarn will have up to five percent residual shrinkage.

6.5. Subject term (key word) listing.

Branch type polyethylene

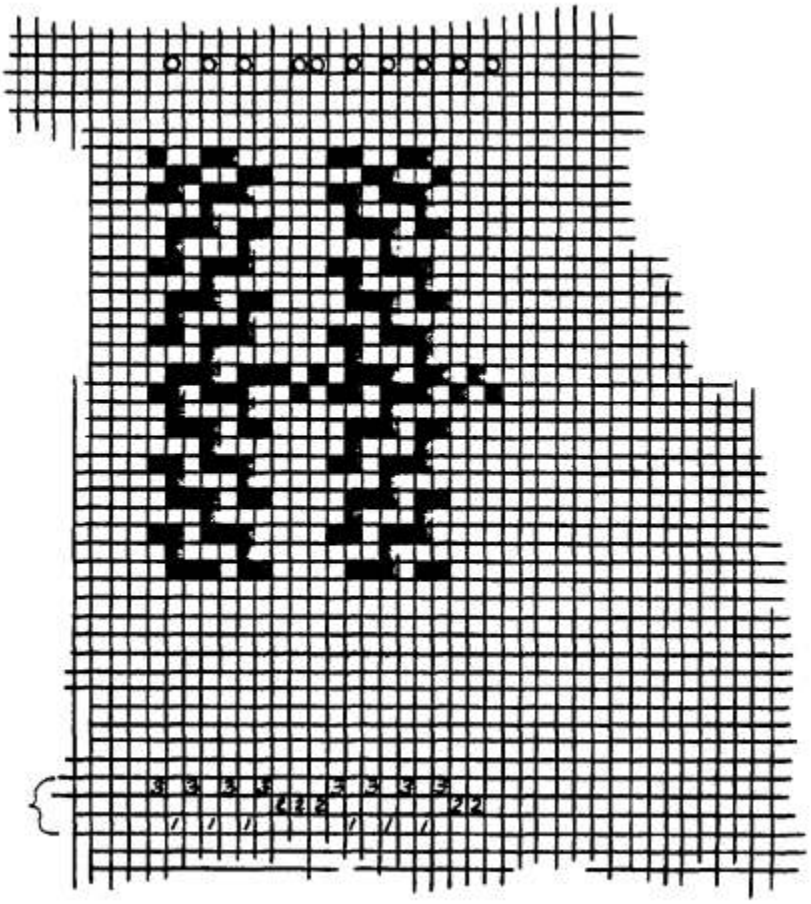
Compressometer

Medium density polyethylene

Spacer Cloth

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TYPE I AND TYPE III



(SHRINKABLE WARP YARNS SHOWN BY CIRCLES)

FIGURE 1. Warp arrangement.

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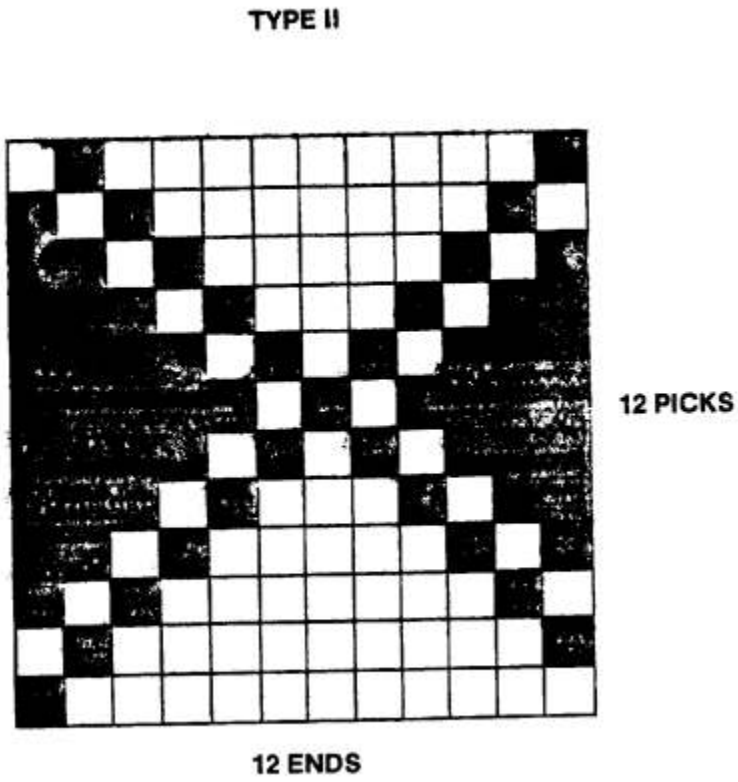


FIGURE 2. Weave diagram.

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CONCLUDING MATERIAL

Custodians:

Army - GL

DLA - CT

Air Force - 11

Preparing activity:

Air Force - 11

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